

Seminar 7: Frequency Response of FIR Filters

Questions and Final Answers

1. Sinusoidal response of a simple FIR filter

Consider the FIR system $y[n] = x[n] + 0.5x[n-1]$ and the complex-exponential input $x[n] = e^{j\hat{\omega}n}$.

(a) Substitute the input into the difference equation and show that the output can be written as $y[n] = H(e^{j\hat{\omega}})e^{j\hat{\omega}n}$.

Answer. $y[n] = (1 + 0.5e^{-j\hat{\omega}})e^{j\hat{\omega}n}$.

(b) Determine $H(e^{j\hat{\omega}})$.

Answer. $H(e^{j\hat{\omega}}) = 1 + 0.5e^{-j\hat{\omega}}$.

(c) Evaluate $H(e^{j\hat{\omega}})$ at $\hat{\omega} = 0, \pi/2$, and π , and give each result in magnitude-phase form.

Answer. $1.5\angle 0$, $1.1180\angle(-0.46365)$ rad, and $0.5\angle 0$.

(d) For the real input $x[n] = 3\cos(\pi n/2 + \pi/4)$, determine the steady-state output.

Answer. $y[n] = 3.3541\cos(\pi n/2 + 0.32175)$.

(e) Explain what is meant by saying that a complex exponential is an eigenfunction of an LTI system.

Answer. The output is the same complex exponential multiplied by $H(e^{j\hat{\omega}})$; its frequency is unchanged.

2. From impulse response to frequency response

An FIR filter has impulse response $h[n] = \delta[n] + 2\delta[n-1] + \delta[n-2]$.

(a) Write the corresponding difference equation.

Answer. $y[n] = x[n] + 2x[n-1] + x[n-2]$.

(b) Derive $H(e^{j\hat{\omega}})$ directly from $h[n]$.

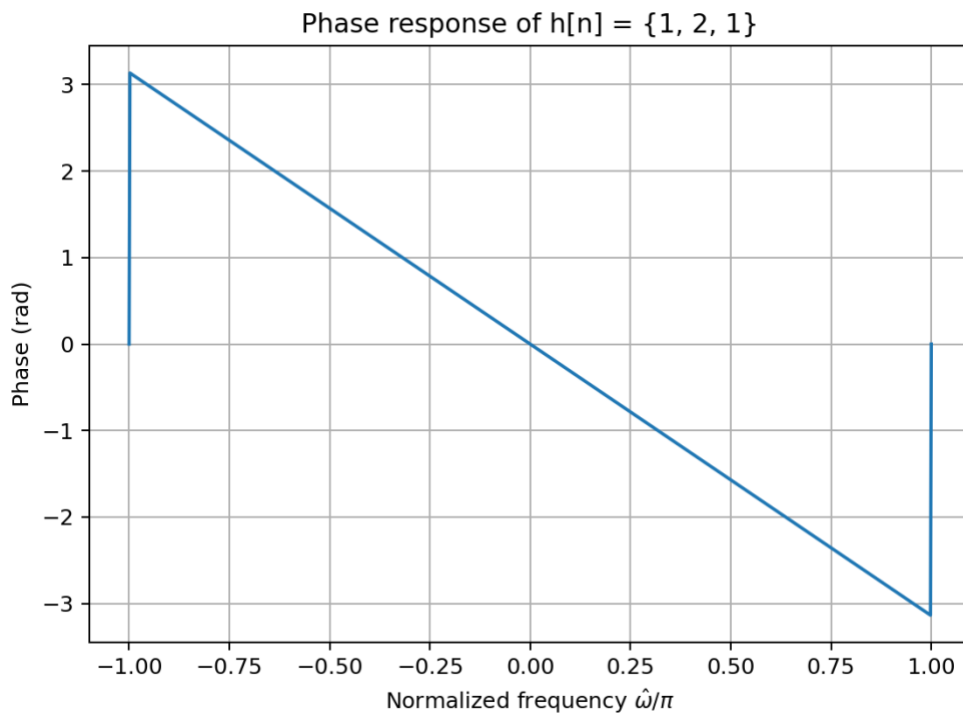
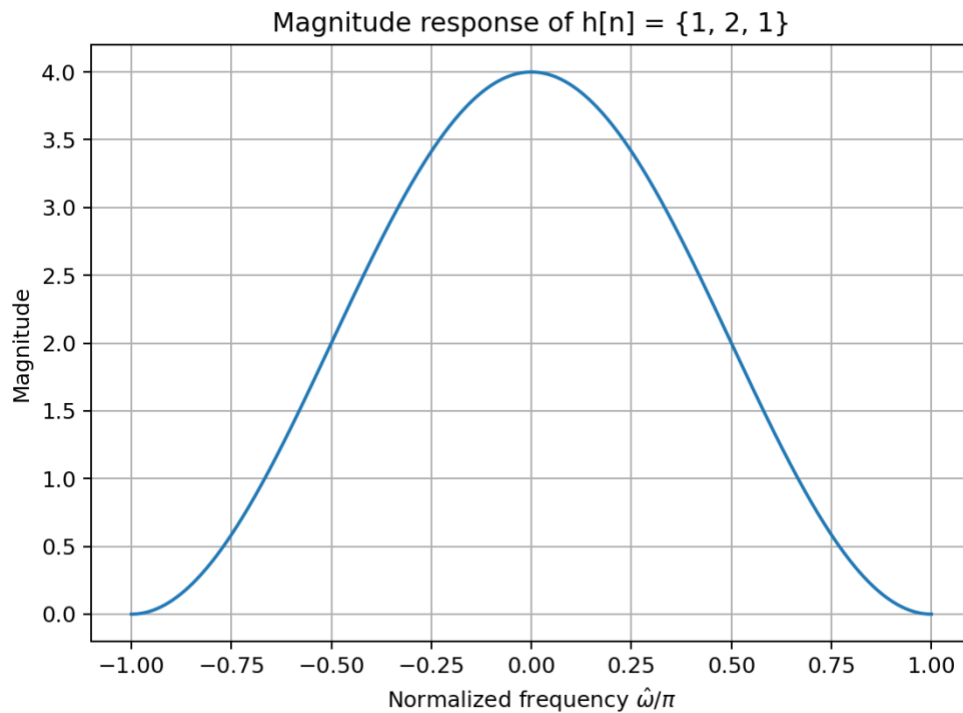
Answer. $H(e^{j\hat{\omega}}) = 1 + 2e^{-j\hat{\omega}} + e^{-j2\hat{\omega}}$.

(c) Factor the frequency response into a real amplitude factor and a linear-phase factor.

Answer. $H = 4\cos^2(\hat{\omega}/2)e^{-j\hat{\omega}}$.

(d) Sketch $|H(e^{j\hat{\omega}})|$ and $\angle H(e^{j\hat{\omega}})$ for $-\pi \leq \hat{\omega} \leq \pi$.

Answer. Magnitude = $4\cos^2(\hat{\omega}/2)$; phase = $-\hat{\omega}$ (except at zeros).



(e) Identify all frequencies in this interval for which the output is zero.

Answer. $\hat{\omega} = \pm\pi$.

(f) State whether the filter behaves qualitatively as a low-pass, high-pass, or neither.

Answer. Low-pass.

3. Periodicity and symmetry of frequency response

For a real-valued FIR filter with frequency response $H(e^{j\hat{\omega}})$:

(a) Prove from the definition that $H(e^{j(\hat{\omega}+2\pi)}) = H(e^{j\hat{\omega}})$.

Answer. $H(e^{j(\hat{\omega}+2\pi)}) = H(e^{j\hat{\omega}})$.

(b) Show that $H(e^{-j\hat{\omega}}) = H^*(e^{j\hat{\omega}})$.

Answer. $H(e^{-j\hat{\omega}}) = H^*(e^{j\hat{\omega}})$.

(c) Deduce the symmetry properties of the magnitude and phase responses.

Answer. $|H(-\hat{\omega})| = |H(\hat{\omega})|$ and $\angle H(-\hat{\omega}) = -\angle H(\hat{\omega})$ modulo 2π .

(d) If $|H(e^{j0.4\pi})| = 0.7$ and $\angle H(e^{j0.4\pi}) = -0.3\pi$, determine the magnitude and phase at -0.4π .

Answer. $|H(e^{-j0.4\pi})| = 0.7$, $\angle H(e^{-j0.4\pi}) = +0.3\pi$.

(e) Explain why it is normally sufficient to plot the response over one 2π interval.

Answer. The frequency response is 2π -periodic.

4. Pure delay filter

Consider the delay system $y[n] = x[n - n_0]$.

(a) Determine its impulse response $h[n]$.

Answer. $h[n] = \delta[n - n_0]$.

(b) Derive the frequency response $H(e^{j\hat{\omega}})$.

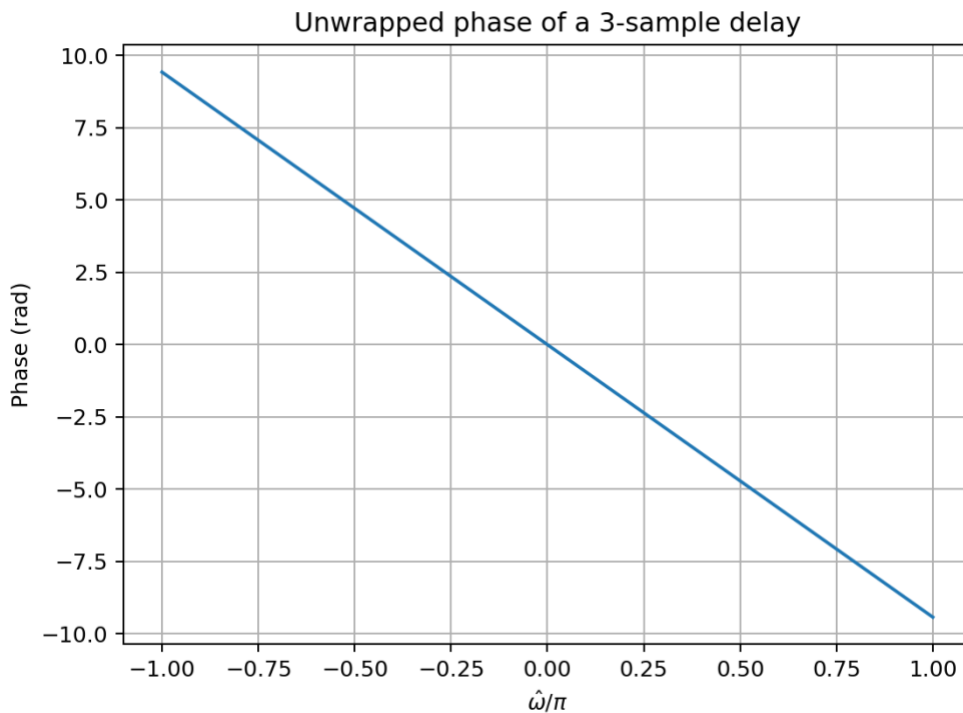
Answer. $H = e^{-j\hat{\omega}n_0}$.

(c) Determine its magnitude and phase responses.

Answer. $|H| = 1$, $\angle H = -\hat{\omega}n_0$.

(d) Sketch the phase response for $n_0 = 3$ over $-\pi \leq \hat{\omega} \leq \pi$.

Answer. Unwrapped phase: $-3\hat{\omega}$.



(e) For $x[n] = 2\cos(0.3\pi n + \pi/6)$, write the output for $n_0 = 3$.

Answer. $y[n] = 2\cos(0.3\pi n - 11\pi/15)$.

(f) Explain why a pure delay changes phase but not amplitude.

Answer. Because $|e^{-j\hat{\omega}n_0}| = 1$ for all $\hat{\omega}$.

5. First-difference filter

Consider the first-difference system $y[n] = x[n] - x[n - 1]$.

(a) Write $h[n]$ and derive $H(e^{j\hat{\omega}})$.

Answer. $h[n] = \delta[n] - \delta[n - 1]$, $H = 1 - e^{-j\hat{\omega}}$.

(b) Show that $|H(e^{j\hat{\omega}})| = 2|\sin(\hat{\omega}/2)|$.

Answer. $|H| = 2|\sin(\hat{\omega}/2)|$.

(c) Determine the phase response over $0 < \hat{\omega} < \pi$.

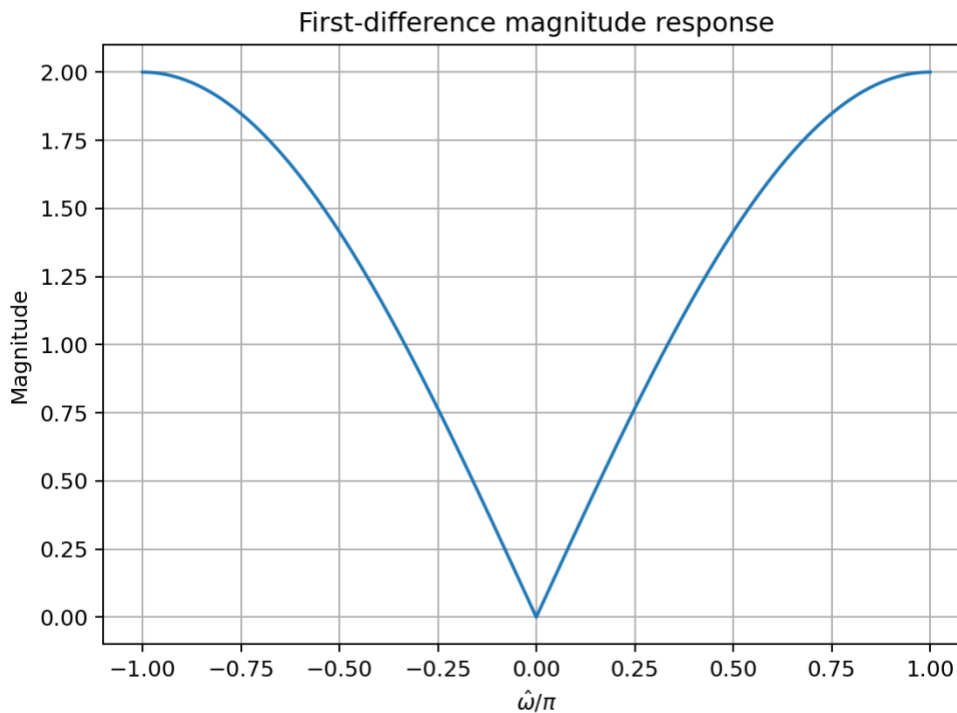
Answer. $\angle H = \pi/2 - \hat{\omega}/2$.

(d) Find all frequencies at which the response is zero.

Answer. $\hat{\omega} = 2\pi k$; in $[-\pi, \pi]$, $\hat{\omega} = 0$.

(e) Evaluate the gains at $\hat{\omega} = 0.1\pi$, 0.5π , and π and explain why this filter acts as a simple high-pass filter.

Answer. Gains: 0.3129, 1.4142, 2; high-pass because gain increases from 0 at DC.



(f) Determine the response to the unit-step input $u[n]$.

Answer. $y[n] = \delta[n]$.

6. Superposition and a multicomponent input

A real FIR filter has frequency-response values $H(e^{j0.1\pi}) = 0.95e^{-j0.2\pi}$, $H(e^{j0.6\pi}) = 0.20e^{-j0.5\pi}$.

The input is $x[n] = 4\cos(0.1\pi n + 0.3\pi) + 2\cos(0.6\pi n - 0.2\pi)$.

(a) Determine the steady-state output using superposition.

Answer. $y[n] = 3.8\cos(0.1\pi n + 0.1\pi) + 0.4\cos(0.6\pi n - 0.7\pi)$.

(b) State the output amplitude and phase of each component.

Answer. $(3.8, 0.1\pi)$ and $(0.4, -0.7\pi)$.

(c) Which input component is attenuated more strongly?

Answer. The 0.6π component.

(d) Explain why no new frequencies are created by an LTI FIR filter.

Answer. LTI filtering scales/phase-shifts existing frequency components; it does not generate new ones.

7. Transient and steady-state response

Consider the order-4 FIR filter $h[n] = \{1, -2, 4, -2, 1\}$, $0 \leq n \leq 4$,

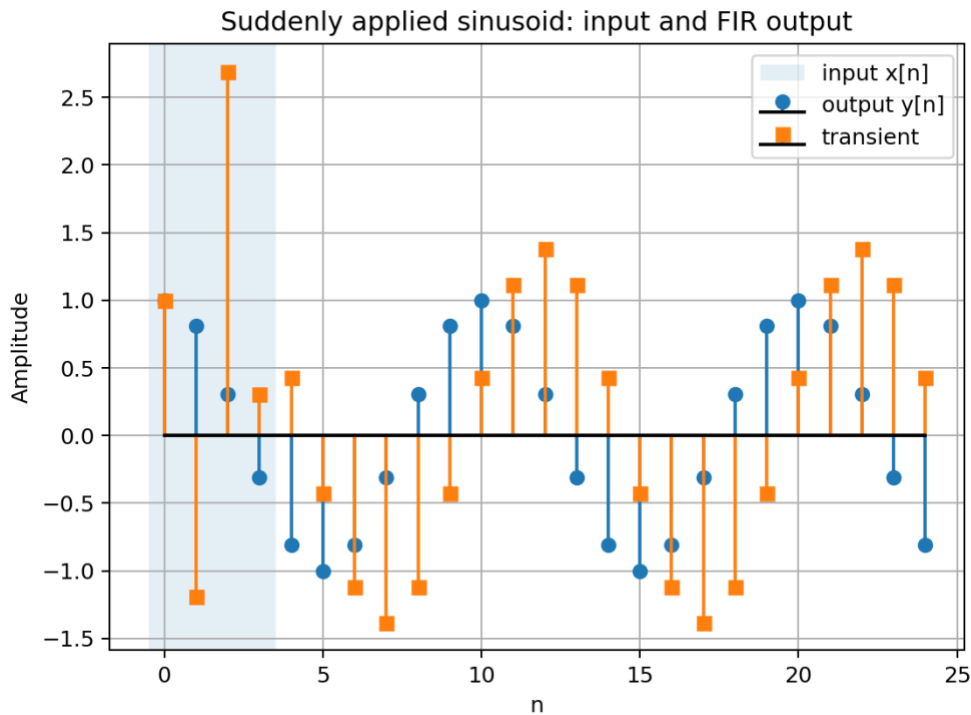
with the suddenly applied input $x[n] = \cos(0.2\pi n)u[n]$.

(a) For which sample indices is the output in the initial transient region?

Answer. $n = 0, 1, 2, 3$.

(b) Compute $y[0]$, $y[1]$, $y[2]$, $y[3]$, and $y[4]$ directly from the convolution sum.

Answer. $y[0] = 1$, $y[1] = -1.190983$, $y[2] = 2.690983$, $y[3] = 0.309017$, $y[4] = 0.427051$.



(c) Evaluate $H(e^{j0.2\pi})$ and write the steady-state sinusoidal output.

Answer. $H(e^{j0.2\pi}) = 1.381966e^{-j0.4\pi}$; $y_{ss}[n] = 1.381966\cos(0.2\pi n - 0.4\pi)$.

(d) From what value of n is the steady-state formula valid?

Answer. $n \geq 4$.

(e) Explain why the sinusoidal-response rule does not describe the first few output samples exactly.

Answer. The suddenly applied sinusoid is not a two-sided eigenfunction during the first four samples; the FIR memory is only partially filled.

8. Recovering $h[n]$ and the difference equation from $H(e^{j\hat{\omega}})$

The frequency response of an FIR filter is $H(e^{j\hat{\omega}}) = 2 - 3e^{-j\hat{\omega}} + e^{-j2\hat{\omega}} + 0.5e^{-j3\hat{\omega}}$.

(a) Determine the impulse response $h[n]$.

Answer. $h[n] = 2\delta[n] - 3\delta[n-1] + \delta[n-2] + 0.5\delta[n-3]$.

(b) Write the FIR difference equation.

Answer. $y[n] = 2x[n] - 3x[n-1] + x[n-2] + 0.5x[n-3]$.

(c) State the filter order and length.

Answer. $M = 3$, $L = 4$.

(d) Evaluate $H(e^{j\hat{\omega}})$ at DC and at the Nyquist frequency $\hat{\omega} = \pi$.

Answer. $H(1) = 0.5$, $H(e^{j\pi}) = 5.5$.

(e) Explain the direct correspondence among the coefficient list, $h[n]$, the difference equation, and $H(e^{j\hat{\omega}})$.

Answer. The FIR coefficients are simultaneously the nonzero samples of $h[n]$, the weights in the difference equation, and the coefficients of $e^{-j\hat{\omega}k}$ in H .

9. Cascaded FIR systems in the frequency domain

Two FIR systems are connected in cascade: $h_1[n] = \{1, 1\}$, $h_2[n] = \{1, -1\}$.

(a) Find $H_1(e^{j\hat{\omega}})$ and $H_2(e^{j\hat{\omega}})$.

Answer. $H_1 = 1 + e^{-j\hat{\omega}}$, $H_2 = 1 - e^{-j\hat{\omega}}$.

(b) Determine the overall frequency response $H(e^{j\hat{\omega}})$.

Answer. $H = 1 - e^{-j2\hat{\omega}}$.

(c) Compute the equivalent impulse response by convolution and verify that it gives the same $H(e^{j\hat{\omega}})$.

Answer. $h[n] = \{1, 0, -1\}$.

(d) Identify the frequencies at which the cascade has zero response.

Answer. $\hat{\omega} = k\pi$; principal interval: $-\pi, 0, \pi$.

(e) Show that reversing the order of the two systems does not change the overall response.

Answer. $H_1H_2 = H_2H_1$ and $h_1 * h_2 = h_2 * h_1$.

10. Frequency response of an L-point moving average

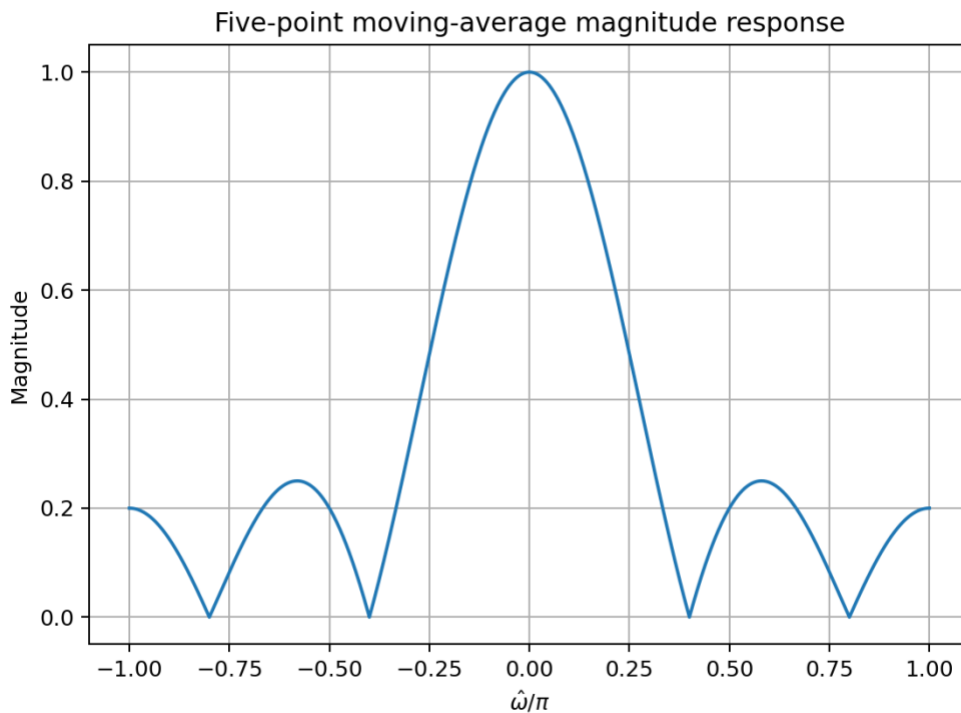
An L -point causal moving-average filter is $y[n] = \frac{1}{L} \sum_{k=0}^{L-1} x[n-k]$.

(a) Derive $H(e^{j\hat{\omega}})$ as a finite geometric series.

Answer. $H = \frac{1}{L} \frac{1 - e^{-jL\hat{\omega}}}{1 - e^{-j\hat{\omega}}}$.

(b) Rewrite it as a product of a real amplitude factor and a linear-phase factor.

Answer. $H = e^{-j(L-1)\hat{\omega}/2} \frac{\sin(L\hat{\omega}/2)}{L\sin(\hat{\omega}/2)}$.



(c) Find the normalized frequencies of all zeros in $0 < \hat{\omega} \leq \pi$.

Answer. $\hat{\omega}_m = 2\pi m/L$ for $m = 1, \dots, \lfloor L/2 \rfloor$.

(d) What is the gain at DC?

Answer. $|H(1)| = 1$.

(e) What delay is represented by the phase factor?

Answer. $(L - 1)/2$ samples.

(f) Explain how increasing L changes smoothing, main-lobe width, the number of zeros, and delay.

Answer. Increasing L gives stronger smoothing, a narrower main lobe, more zeros, and a larger delay $(L - 1)/2$.

11. Comparing even- and odd-length moving averages

Compare a four-point and a five-point causal moving-average filter at normalized frequency $\hat{\omega} = 0.2\pi$.

(a) Compute the magnitude response of each filter.

Answer. $|H_4| = 0.7694$, $|H_5| = 0.6472$.

(b) Determine the phase delay, in samples, of each filter.

Answer. 1.5 samples and 2 samples.

(c) Which filter provides stronger smoothing at this frequency?

Answer. The five-point average.

(d) Which filter has an integer-sample delay? Which has a fractional-sample delay?

Answer. Five-point: 2 samples; four-point: 1.5 samples.

(e) Explain how a 1.5-sample delay should be interpreted after ideal continuous-time reconstruction.

Answer. $1.5T_s$ seconds of continuous-time delay.

12. Using a moving average to cancel a sinusoidal disturbance

A measured sequence contains a slowly varying desired signal plus a sinusoidal disturbance at normalized frequency $\hat{\omega} = \pi/2$.

(a) Find the smallest moving-average length L that has an exact zero at $\hat{\omega} = \pi/2$.

Answer. $L = 4$.

(b) Write the corresponding difference equation.

Answer. $y[n] = \frac{1}{4}(x[n] + x[n-1] + x[n-2] + x[n-3])$.

(c) Determine its delay.

Answer. 1.5 samples.

(d) Would the same filter exactly cancel a disturbance at $\hat{\omega} = 0.4\pi$? Explain.

Answer. No; 0.4π is not a zero of the four-point averager.

(e) Suggest a moving-average length that would exactly cancel a component at $\hat{\omega} = 0.4\pi$, if possible.

Answer. $L = 5$.

13. Filtering a sampled continuous-time signal

A continuous-time signal $x(t) = 2\cos(2\pi 50t) + \cos(2\pi 300t)$

is sampled at $f_s = 1000$ samples/s, filtered by a five-point moving average, and ideally reconstructed.

(a) Determine the normalized radian frequencies of the two sinusoidal components.

Answer. 0.1π and 0.6π rad/sample.

(b) Evaluate the moving-average frequency response at both frequencies.

Answer. $H(0.1\pi) = 0.90403e^{-j0.2\pi}$; $H(0.6\pi) = 0.24721e^{-j0.2\pi}$.

(c) Determine the amplitudes and phase shifts of the two reconstructed sinusoidal components.

Answer. 50 Hz: amplitude 1.80806, phase -0.2π ; 300 Hz: amplitude 0.24721, phase -0.2π .

(d) Write the reconstructed continuous-time output $y(t)$.

Answer. $y(t) = 1.80806\cos(2\pi 50t - 0.2\pi) + 0.24721\cos(2\pi 300t - 0.2\pi)$.

(e) Explain which analogue component is preserved more strongly and why.

Answer. The 50-Hz component; the moving average is low-pass.

(f) State the assumption needed for the analogue frequencies to be preserved through sampling and reconstruction.

Answer. No aliasing: all analogue frequencies must lie below $f_s/2 = 500$ Hz.

14. Interpreting a magnitude and phase response

An FIR filter has the following qualitative properties over $0 \leq \hat{\omega} \leq \pi$: unity gain at DC, a smooth decrease in magnitude, a zero at $\hat{\omega} = 2\pi/3$, and approximately linear phase with slope -2 .

(a) What type of frequency-selective behaviour does this suggest?

Answer. Low-pass behaviour.

(b) What sinusoidal component is completely removed?

Answer. $\hat{\omega} = 2\pi/3$.

(c) What time delay is implied by the phase slope?

Answer. 2 samples.

(d) Predict qualitatively the output for a signal containing components at 0.1π , $2\pi/3$, and 0.9π .

Answer. 0.1π : mostly preserved; $2\pi/3$: cancelled; 0.9π : strongly attenuated; surviving components delayed by about 2 samples.

(e) Explain why magnitude and phase plots together are needed to predict the complete sinusoidal output.

Answer. Magnitude gives gain; phase gives phase shift/delay. Both are needed.

15. Computational investigation with Python

Use NumPy, Matplotlib, and scipy.signal.freqz to investigate the frequency response of FIR filters.

(a) Plot magnitude and phase for the filters $h_1[n] = \{1, -1\}$, $h_2[n] = \{1/3, 1/3, 1/3\}$, and $h_3[n] = \{1, 2, 1\}$.

Answer. Expected: h_1 high-pass/DC zero; h_2 three-point low-pass with zeros at $\pm 2\pi/3$; h_3 low-pass with zero at π .

(b) Verify numerically the 2π periodicity and conjugate-symmetry properties for each real FIR filter.

Answer. $H(\hat{\omega} + 2\pi) = H(\hat{\omega})$ and $H(-\hat{\omega}) = H^*(\hat{\omega})$ to numerical precision.

(c) For each filter, locate any frequency-response zeros and relate them to attenuation of sinusoidal inputs.

Answer. h_1 : 0; h_2 : $\pm 2\pi/3$; h_3 : π (principal interval).

(d) Generate a sum of two sinusoids, filter it with $h_2[n]$, and compare measured output amplitudes with predictions from $H(e^{j\hat{\omega}})$.

Answer. Measured steady-state amplitudes should equal input amplitudes multiplied by $|H(e^{j\hat{\omega}})|$ at each frequency.

(e) Repeat using a suddenly applied sinusoid and identify the transient and steady-state portions of the output.

Answer. For h_2 (order 2), transient: $n = 0, 1$; steady state begins at $n = 2$.

(f) Cascade $h_1[n]$ and $h_2[n]$, and verify in both time and frequency domains that convolution of impulse responses corresponds to multiplication of frequency responses.

Answer. $h_{eq} = \{1/3, 0, 0, -1/3\}$ and $H_{eq} = H_1 H_2 = (1/3)(1 - e^{-j3\hat{\omega}})$.